HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Mathematics

F18CD

Real Analysis

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024

Duration: 2 hours

Answer All Questions

Attached: No additional attachments
Supplied: No additional material supplied
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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1. (a) Write down the definition of the infimum of a set.

[3 marks]

(b) Prove that for each $x, y \in \mathbb{R}$ with $0 < x < y$, there exists a $z \in \mathbb{Q}$ such that $x < z < y$.

[7 marks]

(c) Let A be the set

$$A = \{x \in \mathbb{Q} : 1 \leq x < 3\} \subset \mathbb{R}.$$

Using part (b), or otherwise, prove that the infimum of A is 1 and the supremum of A is 3. Is $\max(A) = 3$ and $\min(A) = 1$? In either case, explain your answer.

[6 marks]

(d) Let $B \subset \mathbb{Z} \subset \mathbb{R}$ such that B is non-empty and bounded below. Does the minimum of B always exist? If so, provide a proof. Otherwise provide a counter-example.

[4 marks]

2. (a) Write down the definition of a convergent sequence (a_n) to converge to $a \in \mathbb{R}$.

[3 marks]

(b) Let (a_n) and (c_n) be convergent sequences with $\lim(a_n) = \lim(c_n) = a$. Prove that if a sequence (b_n) satisfies $a_n \leq b_n \leq c_n$ for all $n > m$, with $m \in \mathbb{N}$, then $\lim(b_n) = a$.

[6 marks]

(c) Using part (b), or otherwise, prove that the sequences (x_n) and (y_n) given by

$$x_n = \frac{\cos(n^3)}{n^5}$$

and

$$y_n = \frac{n!}{n^n}$$

are convergent. Write down the value that they converge to explicitly.

[8 marks]

(d) Suppose that (a_n), (b_n) and (c_n) are sequences with (a_n) convergent, (c_n) divergent and $a_n \leq b_n \leq c_n$ for all $n > m$ for some $m \in \mathbb{N}$. Does it follow that (b_n) is convergent? If so, provide a proof. Otherwise, provide a counterexample.

[3 marks]

3. (a) Write down the definition of an absolutely convergent series.

[3 marks]

(b) Prove that a series $\sum_{k=1}^{\infty} a_k$ is convergent if and only if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for all $n > m > N$.

[6 marks]

(c) Consider the series

$$\sum_{k=1}^{\infty} \frac{3k^2 + \cos^3(k)}{k^7}.$$

Determine if the series is convergent. Make sure to explicitly state any of the series tests used to arrive at your result.

[5 marks]

(d) Prove that the series

$$\sum_{k=1}^{\infty} (-1)^k \sin\left(\frac{1}{k}\right)$$

is convergent. Does it follow that

$$\sum_{k=1}^{\infty} (-1)^k \cos\left(\frac{1}{k}\right)$$

is convergent? Make sure to explain your answer.

[6 marks]

4. (a) Let $f : D \to \mathbb{R}$ where $D \subset \mathbb{R}$. Write down the definition of the function f to be differentiable at a point $c \in D$

[3 marks]

(b) Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$, with $D \subset \mathbb{R}$, both be differentiable at the point $c \in D$. Prove that

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

[6 marks]

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be a real function given by

$$f(x) = \begin{cases} \dfrac{6x^2 - 5x - 6}{2x - 3}, & \text{if } x \neq 3/2 \\ 13/2, & \text{if } x = 3/2 \end{cases}.$$

Prove directly from the definition of a limit of a function that $f(x)$ is continuous at $x = 3/2$.

[7 marks]

(d) Suppose that $f : D \to \mathbb{R}$ is continuous at a point $c \in D$. Does it follow that f is differentiable at $c \in D$? If so, provide a proof. Otherwise, provide a counter example.

[4 marks]